

# Jieke Shi

CS Ph.D. Candidate | Research Engineer  
Singapore Management University

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## RESEARCH INTERESTS

Software Engineering, Program Synthesis, AI System Testing, Code LLMs

I design and implement **automated synthesis**-assisted methods for validating the **safety** of AI-enabled system and enhancing the **efficiency** of large language models serving.

## EDUCATION

Aug 2022 – Dec 2025 **Singapore Management University**, School of Computing and Information Systems  
Ph.D. Candidate in Computer Science

- **29 Conference/Journal Publications**
- **ACM SIGSOFT Distinguished Paper Awards** (ASE'22 nomination, ASE'25 winner, ICSE'26 winner)
- **IEEE Computer Society TCSE Distinguished Paper Award** (SANER 2026)
- **IEEE Computer Society Best Paper Award** (2025)
- **SMU Research Staff Excellence Awards** (2025)
- **SMU SCIS Dean's List Award** (2023 & 2025)
- **SMU Presidential Doctoral Fellowship** (2023)
- **Honorable Mention Award** (ACSAC 2022)

Supervisor: ACM/IEEE/ASE Fellow, OUB Chair Professor [David Lo](#)

Aug 2016 – Jun 2020 **Yangzhou University**, School of Information Engineering  
B.Eng. in Electronic Engineering

- **Outstanding Undergraduate Thesis Award** (Top 0.2% in province)
- **Outstanding Graduate** (Top 5%)

## EMPLOYMENT

Jun 2021 – Present **Singapore Management University**, Research Engineer

Working as lead research staff under the AI.SG Programme on the testing of Reinforcement Learning (RL)-enabled systems. Led to multiple publications covering runtime shielding and RL system safety validation, including [J.5, J.7, C.5, C.8, C.9].

Jul 2020 – Apr 2021 **Chinese Academy of Sciences, Institute of Computing**, Research Assistant Intern

Developed a prototype system for generating faithful natural language text from structured data.

## PUBLICATIONS

Note: In software engineering, ICSE and ASE are recognized as top-tier conferences, and TOSEM and TSE are top-tier journals (CORE A\*); MSR, ICSME, and SANER are prominent conferences covering software mining, analysis, and evolution (CORE A). In security, S&P is a top-tier conference (CORE A\*), while ACSAC is a leading venue with a focus on system security applications (CORE A).

[C]: Full Conference Paper (Total: **15**)   [J]: Full Journal Paper (Total: **8**)   [S]: Short Paper (Total: **6**)

[2026](#)

- [S.6] Chengyan Ma, **Jieke Shi**, Ruidong Han, Ye Liu, Yuqing Niu, and David Lo. “*Finding Missing Input Validation in TEEs via LLM-Assisted Symbolic Execution*”. In Proceedings of the 3rd ACM International Conference on AI Foundation Models and Software Engineering (**FORGE 2026**, New Idea Papers).
- [J.8] Junda He, **Jieke Shi**, Terry Yue Zhuo, Christoph Treude, Jiamou Sun, Zhenchang Xing, Xiaoning Du, and David Lo. “*LLM-as-a-Judge for Software Engineering: Literature Review, Vision, and the Road Ahead*”. ACM Transactions on Software Engineering and Methodology (**TOSEM 2026**). [\[PDF\]](#)
- [C.16] Yuqing Niu, **Jieke Shi**, Ruidong Han, Ye Liu, Chengyan Ma, Yunbo Lyu, and David Lo. “*What You Trust Is Insecure: Demystifying How Developers (Mis)Use Trusted Execution Environments in Practice*”. In Proceedings of the 33rd IEEE International Conference on Software Analysis, Evolution and Reengineering (**SANER 2026**, **IEEE Computer Society TCSE Distinguished Paper Award**). [\[PDF\]](#) [\[Code\]](#)
- [S.5] Huihui Huang, **Jieke Shi**, Junkai Chen, Ting Zhang, Yikun Li, Chengran Yang, Eng Lieh Ouh, Lwin Khin Shar, and David Lo. “*PenForge: On-the-Fly Expert Agent Construction for Automated Penetration Testing*”. In Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (**ICSE 2026**, NIER Track). [\[PDF\]](#) [\[Code\]](#)
- [C.14] Huihui Huang, Ratnadira Widyasari, Ting Zhang, Ivana Clairine IRSAN, **Jieke Shi**, Han Wei Ang, Frank Liauw, Eng Lieh Ouh, Lwin Khin Shar, Hong Jin Kang, and David Lo. “*Back to the Basics: Rethinking Issue-Commit Linking with LLM-Assisted Retrieval*”. In Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (**ICSE 2026**). [\[PDF\]](#) [\[Code\]](#)
- [C.13] Chengran Yang, Zhensu Sun, Hong Jin Kang, **Jieke Shi**, and David Lo. “*Think Like Human Developers: Harnessing Community Knowledge for Structured Code Reasoning*”. In Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (**ICSE 2026**, **ACM SIGSOFT Distinguished Paper Award**). [\[PDF\]](#)

## 2025

- [J.7] **Jieke Shi**, Zhou Yang, Junda He, Bowen Xu, Dongsun Kim, DongGyun Han, and David Lo. “*Finding Safety Violations of AI-Enabled Control Systems through the Lens of Synthesized Proxy Programs*”. ACM Transactions on Software Engineering and Methodology (**TOSEM 2025**). [\[PDF\]](#) [\[Code\]](#)
- [J.6] **Jieke Shi**, Zhou Yang, and David Lo. “*Efficient and Green Large Language Models for Software Engineering: Literature Review, Vision, and the Road Ahead*”. ACM Transactions on Software Engineering and Methodology (**TOSEM 2025**). [\[PDF\]](#)
- [J.5] **Jieke Shi**, Junda He, Zhou Yang, Đorđe Žikelić, and David Lo. “*Synthesizing Efficient and Permissive Programmatic Runtime Shields for Neural Policies*”. ACM Transactions on Software Engineering and Methodology (**TOSEM 2025**). [\[PDF\]](#)
- [J.4] Zhou Yang, **Jieke Shi**, Premkumar Devanbu, and David Lo. “*Ecosystem of Large Language Models for Code*”. ACM Transactions on Software Engineering and Methodology (**TOSEM 2025**). [\[PDF\]](#)
- [C.12] Yunbo Lyu, Zhou Yang, **Jieke Shi**, Jianming Chang, Yue Liu, and David Lo. “*“My productivity is boosted, but ...” Demystifying Users’ Perception on AI Coding Assistants*”. In Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (**ASE 2025**, **ACM SIGSOFT Distinguished Paper Award**). [\[PDF\]](#) [\[Code\]](#)

## 2024

- [C.11] **Jieke Shi**, Zhou Yang, Hong Jin Kang, Bowen Xu, Junda He, and David Lo. “*Greening Large Language Models of Code*”. In Proceedings of the 46th IEEE/ACM International Conference on Software Engineering (**ICSE 2024**). [\[PDF\]](#) [\[Code\]](#)

- [J.3] Zhou Yang, **Jieke Shi**, Muhammad Hilmi Asyrofi, Bowen Xu, Xin Zhou, DongGyun Han, and David Lo. “Prioritizing Speech Test Cases”. ACM Transactions on Software Engineering and Methodology (**TOSEM 2024**). [[PDF](#)] [[Code](#)]
- [J.2] Zhou Yang, Bowen Xu, Jie M. Zhang, Hong Jin Kang, **Jieke Shi**, Junda He, and David Lo. “Stealthy Backdoor Attack for Code Models”. IEEE Transactions on Software Engineering (**TSE 2024**, 🏆 **IEEE Computer Society Best Paper Award**). [[PDF](#)] [[Code](#)]
- [J.1] Zhou Yang, Zhipeng Zhao, Chenyu Wang, **Jieke Shi**, Dongsun Kim, DongGyun Han, and David Lo. “Gotcha! This Model Uses My Code! Evaluating Membership Leakage Risks in Code Models”. IEEE Transactions on Software Engineering (**TSE 2024**). [[PDF](#)]
- [C.10] Chen Gong, Zhou Yang, Yunpeng Bai, Junda He, **Jieke Shi**, Kecen Li, Arunesh Sinha, Bowen Xu, Xinwen Hou, David Lo, and Tianhao Wang. “BAFFLE: Hiding Backdoors in Offline Reinforcement Learning Datasets”. In Proceedings of the 45th IEEE Symposium on Security and Privacy (**S&P 2024**). [[PDF](#)] [[Code](#)]
- [C.9] Junda He, Zhou Yang, **Jieke Shi**, Chengran Yang, Kisub Kim, Bowen Xu, Xin Zhou, and David Lo. “Curiosity-Driven Testing for Sequential Decision-Making Process”. In Proceedings of the 46th IEEE/ACM International Conference on Software Engineering (**ICSE 2024**). [[PDF](#)] [[Code](#)]
- [C.8] Zhou Yang, Zhipeng Zhao, Chenyu Wang, **Jieke Shi**, Dongsun Kim, DongGyun Han, and David Lo. “Unveiling Memorization in Code Models”. In Proceedings of the 46th IEEE/ACM International Conference on Software Engineering (**ICSE 2024**). [[PDF](#)] [[Code](#)]

#### 2023

- [C.7] Zhou Yang, Chenyu Wang, **Jieke Shi**, Thong Hoang, Pavneet Kochhar, Qinghua Lu, Zhenchang Xing, and David Lo. “What Do Users Ask in Open-Source AI Repositories? An Empirical Study of GitHub Issues”. In Proceedings of the 20th IEEE/ACM International Conference on Mining Software Repositories (**MSR 2023**). [[PDF](#)] [[Code](#)]

#### 2022

- [C.6] **Jieke Shi**, Zhou Yang, Bowen Xu, Hong Jin Kang, and David Lo. “Compressing Pre-trained Models of Code into 3 MB”. In Proceedings of the 37th IEEE/ACM International Conference on Automated Software Engineering (**ASE 2022**, 🌟 **ACM SIGSOFT Distinguished Paper Award Nomination**). [[PDF](#)] [[Poster](#)] [[Code](#)]
- [S.2] **Jieke Shi**, Zhou Yang, Junda He, Bowen Xu, and David Lo. “Can Identifier Splitting Improve Open-Vocabulary Language Model of Code?”. In Proceedings of the 29th IEEE International Conference on Software Analysis, Evolution and Reengineering (**SANER 2022**, ERA Track). [[PDF](#)] [[Poster](#)] [[Code](#)]
- [C.5] Chen Gong, Zhou Yang, Yunpeng Bai, **Jieke Shi**, Arunesh Sinha, Bowen Xu, David Lo, Xinwen Hou, Guoliang Fan. “Curiosity-Driven and Victim-Aware Adversarial Policies”. In Proceedings of the 39th Annual Computer Security Applications Conference (**ACSAC 2022**, 🏆 **Honorable Mention Award**). [[PDF](#)] [[Code](#)]
- [C.4] Chengran Yang, Bowen Xu, Ferdian Thung, Yucen Shi, Ting Zhang, Zhou Yang, Xin Zhou, **Jieke Shi**, Junda He, DongGyun Han, and David Lo. “Answer Summarization for Technical Queries: Benchmark and New Approach”. In Proceedings of the 37th IEEE/ACM International Conference on Automated Software Engineering (**ASE 2022**). [[PDF](#)] [[Code](#)]
- [C.3] Zhou Yang, **Jieke Shi**, Muhammad Hilmi Asyrofi, and David Lo. “Revisiting Neuron Coverage Metrics and Quality of Deep Neural Networks”. In Proceedings of the 29th IEEE International Conference on Software Analysis, Evolution and Reengineering (**SANER 2022**). [[PDF](#)] [[Code](#)]
- [C.2] Ratnadira Widayarsi, Stefanus Agus Haryono, Ferdian Thung, **Jieke Shi**, Constance Tan, Fiona Wee, Jack Phan, and David Lo. “On the Influence of Biases in Bug Localization: Evaluation and Benchmark”. In Proceedings of the 29th IEEE International Conference on Software Analysis, Evolution and Reengineering (**SANER 2022**). [[PDF](#)] [[Code](#)] [[Dataset](#)]

- [C.1] Zhou Yang, **Jieke Shi**, Junda He, and David Lo. “*Natural Attack for Pre-trained Models of Code*”. In Proceedings of the 44th IEEE/ACM International Conference on Software Engineering (ICSE 2022). [[PDF](#)] [[Code](#)]

2021

- [S.1] Zhou Yang , **Jieke Shi** , Shaowei Wang, and David Lo. “*IncBL: Incremental Bug Localization*”. In Proceedings of the 36th IEEE/ACM International Conference on Automated Software Engineering (ASE 2021, Tool Demonstrations). [[PDF](#)] [[Poster](#)] [[Code](#)]
- [S.2] Zhou Yang, Harshit Jain, **Jieke Shi**, Muhammad Hilmi Asyrofi, and David Lo. “*BiasHeal: On-the-Fly Black-Box Healing of Bias in Sentiment Analysis Systems*”. In Proceedings of the 37th IEEE International Conference on Software Maintenance and Evolution (ICSME 2021, NIER Track). [[PDF](#)] [[Code](#)]
- [S.3] Muhammad Hilmi Asyrofi, Zhou Yang, **Jieke Shi**, Chu Wei Quan, and David Lo. “*Can Differential Testing Improve Automatic Speech Recognition Systems?*”. In Proceedings of the 37th IEEE International Conference on Software Maintenance and Evolution (ICSME 2021, NIER Track). [[PDF](#)] [[Code](#)]

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## HONORS & AWARDS

- 2026 **ACM SIGSOFT Distinguished Paper Award**, ICSE 2026  
**IEEE Computer Society TCSE Distinguished Paper Awards**, SANER 2026
- 2025 **ACM SIGSOFT Distinguished Paper Award**, ASE 2025  
**IEEE Computer Society Best Paper Award**, IEEE Transactions on Software Engineering  
**OpenAI Researcher Access Program Grant** (\$5,000)  
**Google Cloud Research Credits Grant** (\$2,500)  
**Research Staff Excellence Awards** (awarded to only 4 researchers university-wide), Singapore Management University  
**SMU SCIS PhD Research Excellence Award**, Singapore Management University  
**SMU SCIS Dean’s List Award**, Singapore Management University
- 2023 **Presidential Doctoral Fellowship**, Singapore Management University  
**SCIS Dean’s List Award**, Singapore Management University
- 2022 **Honorable Mention Award**, ACSAC 2022  
**ACM SIGSOFT Distinguished Paper Award Nomination**, ASE 2022
- 2020 **Outstanding Undergraduate Thesis** (Top 0.2% in province), Jiangsu Province, China  
**Outstanding Graduate** (Top 5%), Yangzhou University
- 2019 & Before **National Encouragement Scholarship** (Top 3%), Ministry of Education of China  
**2nd Class Presidential Scholarship**, Yangzhou University  
**1st Class Presidential Scholarship**, Yangzhou University

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## ACADEMIC SERVICES

Program Committee (PC)/Reviewer

- 2026 **International Conference on the Foundations of Software Engineering** (FSE), PC in Ideas, Visions and Reflections Track
- 2026 **Workshop on Human-Centered AI for Software Engineering** (HumanAISE), PC
- 2026 **International Conference on AI-powered Software** (Alware), PC in Main Track
- 2026 **International Conference on Software Maintenance and Evolution** (ICSME), PC in Research Papers & Visions and Emerging Results Track

- 2025–2026 **International Conference on Software Engineering (ICSE)**, Shadow PC
- 2024–2026 **Mining Software Repositories Conference (MSR)**, Junior PC
- 2025–2026 **International Workshop on Green and Sustainable Software (GREENS@ICSE)**, PC
- 2026 **European Conference on Computer Systems (EuroSys)**, Shadow PC & Artifact Evaluation
- 2026 **International Conference on Evaluation and Assessment in Software Engineering (EASE)**, PC in AI Models/Data Track
- 2025 **Annual Meeting of the Association for Computational Linguistics (ACL)**, Reviewer
- 2025 **IEEE International Computer Software and Applications Conference (COMPSAC)**, PC

### Organizing Committee

- 2024 **International Workshop on Software Engineering and AI for Data Quality in Cyber-Physical Systems (SEA4DQ@FSE)**, Web Chair

### Journal Reviewer

- 2025–Present **IEEE Transactions on Dependable and Secure Computing (TDSC)**
- 2025–Present **Empirical Software Engineering (EMSE)**
- 2023–Present **ACM Transactions on Software Engineering and Methodology (TOSEM)**
- 2023–Present **IEEE Transactions on Software Engineering (TSE)**
- 2021–Present **Neurocomputing (Software Section)**

### External Reviewer

- 2021–2025 **International Conference on Software Engineering (ICSE)**
- 2022–2026 **IEEE/ACM International Conference on Automated Software Engineering (ASE)**
- 2023–2025 **ACM International Conference on the Foundations of Software Engineering (ESEC/FSE)**
- 2023 **International Conference on Software Maintenance and Evolution (ICSME)**
- 2023 **International Workshop on LLMs for Code (LLM4Code@ICSE)**
- 2022–2023 **International Joint Conference on Artificial Intelligence (IJCAI)**
- 2022 **IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)**
- 2022 **International Conference on AI Engineering – Software Engineering for AI (CAIN)**
- 2022 **International Workshop on Software Engineering for Responsible AI (SE4RAI@ICSE)**
- 2022 **International Conference on Engineering of Complex Computer Systems (ICECCS)**

## INVITED TALKS

- Dec 2025 **Shanghai Jiao Tong University**, Research Seminar  
*Optimizing and Fortifying AI Software through the Lens of Artifact Synthesis*
- Nov 2025 **A-Mobile Workshop of ASE 2025**, Keynote, Invited by Guozhu Meng  
*Optimizing and Fortifying AI Software through the Lens of Artifact Synthesis*
- Nov 2025 **Korea University**, Workshop on the Next Generation of CodeLLMs and Their Applications, Invited by Dongsun Kim  
*Optimizing and Fortifying AI Software via Program Synthesis*
- Sep 2025 **Singapore Management University**, RISE-WASP Workshop, Invited by Lingxiao Jiang  
*Fortifying Autonomous Systems via Program Synthesis*
- Nov 2023 **Singapore Management University**, Research Cluster Seminar, Invited by David Lo  
*Towards Compact and Efficient Language Models of Code via Compression*



Oct 2023 **North Carolina State University**, Guest Lecture, Invited by Bowen Xu  
*Enhancing Efficiency of Large Language Models of Code via Model Compression* [[Link](#)]

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## TEACHING EXPERIENCE

Jan 2026 **Singapore Management University**, Teaching Assistant in CS706, Invited by David Lo  
*Software Mining and Analysis*, Graduate Core Course

Oct 2023 **North Carolina State University**, Guest Lecture in CSC791, Invited by Bowen Xu  
*Enhancing Efficiency of Large Language Models of Code via Model Compression* [[Link](#)]

Jan 2016 - Jun 2016 **Yangzhou University**, Teaching Assistant  
*Digital Signal Processing*, Undergraduate Core Course

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## GRANT PROPOSAL WRITING EXPERIENCE

2025 - 2027 **ESG-based Responsible AI: Toward Green, Secure, and Compliant LLM Utilisation for Digital Service Development Process**  
CSIRO - A\*STAR Partnership Program, Singapore & Australia  
Total Award: Over SGD \$500K

2024 - 2028 **TrustedSEERs: Trusted Intelligent Work Bots for Engineering Better Software**  
National Research Foundation (NRF), Singapore  
Total Award: Over SGD \$3.2M

2023 - 2025 **TitanCA: Vulnerability Detection using Large Code Models**  
Government Technology Agency (GovTech), Singapore  
Total Award: Over SGD \$3.8M

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## REFEREES

**David Lo**, ACM/IEEE/ASE Fellow, ACM Distinguished Speaker, OUB Chair Professor  
School of Computing and Information Systems, Singapore Management University  
[davidlo@smu.edu.sg](mailto:davidlo@smu.edu.sg)

**Zhou Yang**, Canada CIFAR AI Chair Nominee, Amii Fellow, Assistant Professor  
Department of Computer Science, University of Alberta  
[zy25@ualberta.ca](mailto:zy25@ualberta.ca)